Justification for Publication as a Letter

Gravity Field Dynamics: Field Closure of the Tetrad Frame

The affine connection on the tetrad frame has $K^3 = 64$ independent components. General relativity's torsion-free condition removes 24 of them, retaining 40 Christoffel symbols. The present letter, together with its two companions (CQG-115428 and CQG-115447), identifies the 24 removed components as the second root of the quadratic tetrad equation, determines their field fraction from exchange symmetry, and restores them to recover the full 64-component connection.

The work merits consideration as a Letter for the following reasons.

A new result in connection geometry.

The 40/24 split of the affine connection is shown to be reproduced by the coupling polynomial $f(K, s) = 1 + sK + K^2$ evaluated at its two parity states. The polynomial was derived independently from the rank structure of the tetrad equation in the companion letter. Two independent paths, combinatorial counting and polynomial evaluation, converge on the same numbers. To our knowledge, this concordance has not previously appeared in the literature.

Recovery of the radial acceleration relation with no free parameters.

The constitutive law $\mu(x) = x/(x+1)$, derived from exchange symmetry in the companion letter, reproduces the radial acceleration relation established empirically by McGaugh, Lelli & Schombert (2016) across 175 galaxies with 0.057 dex scatter. The TetB field fraction $1/(x+1)$ is bounded above by unity, which removes the central divergence responsible for the cusp-core tension in NFW profiles without requiring baryonic feedback or additional matter-sector content.

A covariant completion with testable constraints.

Integration of the constitutive law yields a unique kinetic functional whose three terms correspond, via polynomial long division, to the three structural levels of the coupling polynomial. The resulting covariant action preserves $\gamma = 1$ at all accelerations (because the modification enters through the connection rather than as a separate scalar field), satisfies the GW170817 constraint on gravitational-wave speed, and contains no free functions. The Bekenstein lensing deficit familiar from scalar-tensor formulations of MOND is identified as a decomposition artefact that does not arise in the connection-level formulation.

Relevance to the journal's readership.

The result connects tetrad geometry, torsion theory, the AQUAL formulation, and the cold dark matter phenomenology through a single algebraic chain. It is relevant to researchers working on tetrad and Poincaré gauge formulations, relativistic alternatives to dark matter, and observational tests of gravity at galactic scales. All claims are supported by a supplementary covariant-completion note and a self-contained computational verification script.

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